H

NATIONAL JUNIOR COLLEGE

Higher 1

CANDIDATE NAME

BIOLOGY CLASS

REGISTRATION NO.

BIOLOGY 8875/01

JC2 Preliminary Examinations **11 Sept 2008**
Paper 1
Additional materials: Optical Answer Sheet **1 Hour**

READ THIS INSTRUCTIONS FIRST

Write your name and registration number (e.g. 05IP0409 or 07S0620) and Biology Class (e.g. 2bi2E) clearly on the cover page.

Write your Name, Registration No. and Biology Class on your Optical Answer Sheet (OAS).

There are forty questions on this paper. Answer all questions.

For each question there are four possible answers A, B, C and D. Choose the one you consider correct and record your choice in 2B soft pencil on the Optical Answer Sheet.

Each correct answer will score one mark. A mark will NOT be deducted for a wrong answer.

Any rough working should be done in this booklet.

This paper consists of **14** printed pages including cover page.

1 Which of the following best accounts for the difference in structure between amylose which is helical and cellulose which is fibrous?

A Whether alternate subunits are rotated 180^o or not.

B Presence or absence of branching in the macromolecule

C Whether the subunits are monosaccharides or amino acids.

D Different tendency of macromolecule to form hydrogen bonds with water.

2 This is a molecule that is found in various parts of the human body, and is helical in structure as can be observed from the diagram below.

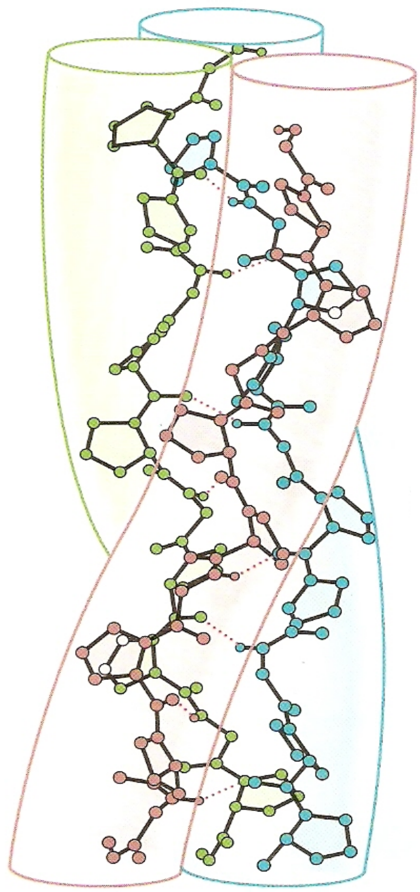


Diagram from:
"Biochemistry" by Garrett and Grisham

What else is true of this molecule?

A Complementary base pairing occurs through the formation of hydrogen bonds between chains.

B Hydrogen bonds within each chain give each chain its helical structure.

C Glutamine occurs very frequently in the sequence of the subunits in the molecule.

D The entire molecule interacts with proteins to form complexes necessary for the control of cell activities.

3 The diagram below shows the synthesis of a new strand of DNA during interphase.

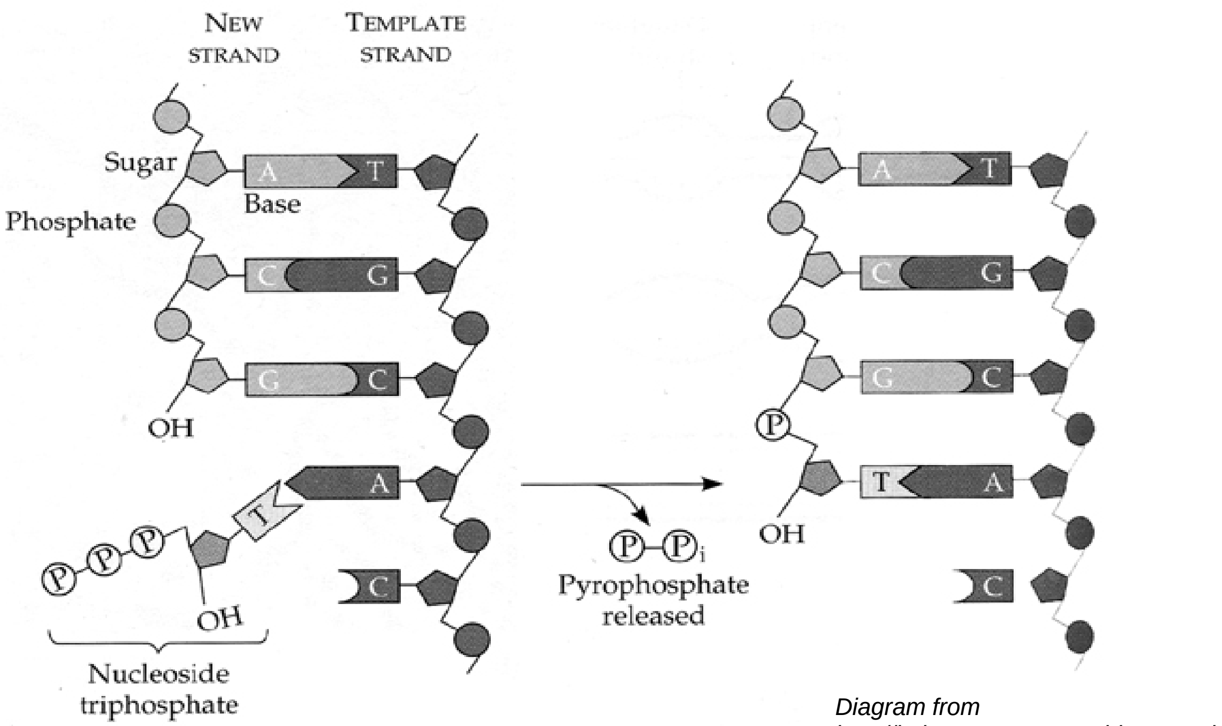


Diagram from http://hshgp.genome.washington.edu

Which of the following shows the correct combination of bond(s) that need to be formed and the kind of reaction that is involved in order for the nucleotide to be added to the DNA chain?

	Bond(s) to be formed	**Reaction(s) involved**
A	Phosphodiester	Condensation and Dephosphorylation
B	Phosphoester	Hydrolysis
C	Phosphoester and Hydrogen	Condensation and Hydrolysis
D	Phosphoester	Dephosphorylation

4 Which of the following is / are true of triglycerides and phospholipids?

i Both contain glycerol

ii Both are amphipathic

iii Triglycerides have a higher carbon content than phospholipids for storage purposes

iv Phospholipids have hydrophilic regions to interact with cell cytoplasm unlike triglycerides

v Triglycerides are formed from glycerol and fatty acids but phospholipids are formed from glycerol and phosphoric acid.

A All of the above

B i, ii and iii

C i, iii and iv

D ii, iii and iv

5 A mitochondrion in a liver cell measured 1.2 μm. On an electron micrograph the length of this organelle was 36mm. What is the magnification of the electron micrograph?

A 3000 times

B 12 000 times

C 15000 times

D 30000 times

6 Which of the following statements about the cell is true?

A Endocytosis is an active process because energy is required to move materials into the cell against their concentration gradients.

B In simple diffusion, materials enter the cell down their concentration gradients whereas facilitated diffusion can move materials against their concentration gradients.

C In osmosis, water, ions and small polar molecules enter the cell by diffusing through a selectively permeable membrane down their concentration gradients.

D Materials such as ions and small polar molecules can enter the cell against their concentration gradient by active transport.

7 Which of the following statement about enzymes is inaccurate?

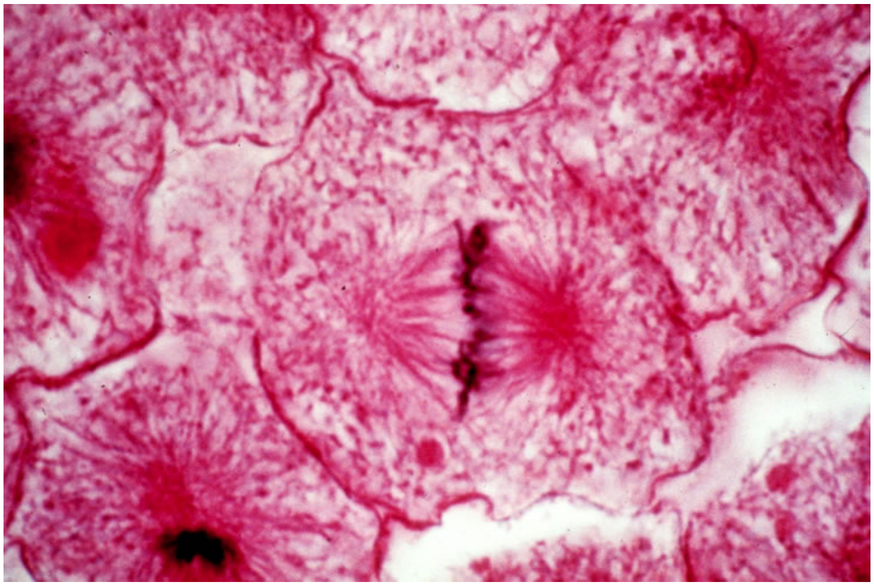
A Enzyme function is not influenced by factors such as lead chloride.

B Enzyme function is reduced if the three-dimensional structure or conformation of an enzyme is altered.

C Enzymes increase the rate of chemical reaction by lowering activation energy barriers.

D Enzymes may require a non-protein cofactor or ion for catalysis to take place.

8 The image below shows a cell division stage of a cell (middle of the diagram) in the white fish blastula.



Choose the most appropriate description that depicts what was happening.

A Attachment of spindle fibre, chromosome fully condensed, separation of sister chromatids

B Chromosomes line up, attachment to centrioles, furrowing of cell membrane

C Chromosomes positioned in the centre, to be pulled towards two nuclei, furrowing of cell membrane

D Chromosome fully condensed, line up in the plate of the cell, asters formed

9 Two parents have a son who has blood group A and phenylketonuria. One parent has blood group O and the other has blood group AB. Neither parent has phenylketonuria.

What is the probability that the second child of these parents will be a girl with blood group B who does not have phenylketonuria ?

A 1 in 16 **B** 1 in 8 **C** 3 in 16 **D** 3 in 8

10 The genotypes of a tomato variety were **TtHh**, **Tthh**, **ttHh** and **tthh**, where **T** = tall and **H** = hairy. Which one of the following crosses would produce progeny giving a phenotypic ratio approximating to 1:1:1:1?

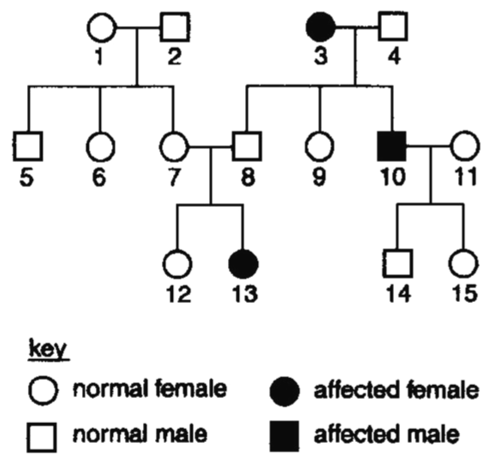
A **TtHh x TtHh**

B **TtHh x Tthh**

C **TtHh x tthh**

D **TTHH x tthh**

11 The family tree shows the inheritance of a condition caused by the recessive allele **r**.



Which of the females are certain to have the genotype **Rr**?

A 1, 6 and 7 **B** 1, 7 and 12 **C** 7, 9 and 15 **D** 9, 12 and 15

12 What type of natural selection is occuring when the average phenotype is selected for and the extreme phenotypes are selected against?

A disruptive

B directional

C stabilising

D reductive

13 Which one of the following statements is related to the principle of adaptive radiation in evolution?

A Unrelated organisms living in similar environments may tend to resemble each other.

B Related organisms living in different environments show modifications of an underlying unity of pattern.

C Selection in a constant environment tends to maintain the most frequently occurring variety of a species.

D Species evolve by selection altering the gene frequencies in interbreeding populations.

14 What do chromosomal aberrations and gene mutations have in common and how are they different?

	Similarity	**Difference**
A	Both may involve addition of nucleotides.	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both may not result in disorders.	Gene mutations do not involve inversions but inversion of segments of chromosomes do occur.
C	Both affect DNA sequence.	Gene mutations occur within a chromosome but chromosomal aberrations may occur across chromosomes.
D	Both may not result in a difference in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

15 Which of the functions of RNA is incorrectly matched?

A Messenger RNAs encode information for synthesis of polypeptides.

B Ribosomal RNAs bind with tRNA to catalyse the formation of phosphodiester bonds.

C Small nuclear RNAs bind with ribonucleoproteins to form spliceosomes.

D Transfer RNAs bind with mRNA to facilitate translation.

16 The Meselson-Stahl experiment is shown below. This experiment was done to prove that replication is via a semi-conservative mode rather than a conservative or dispersive mode.

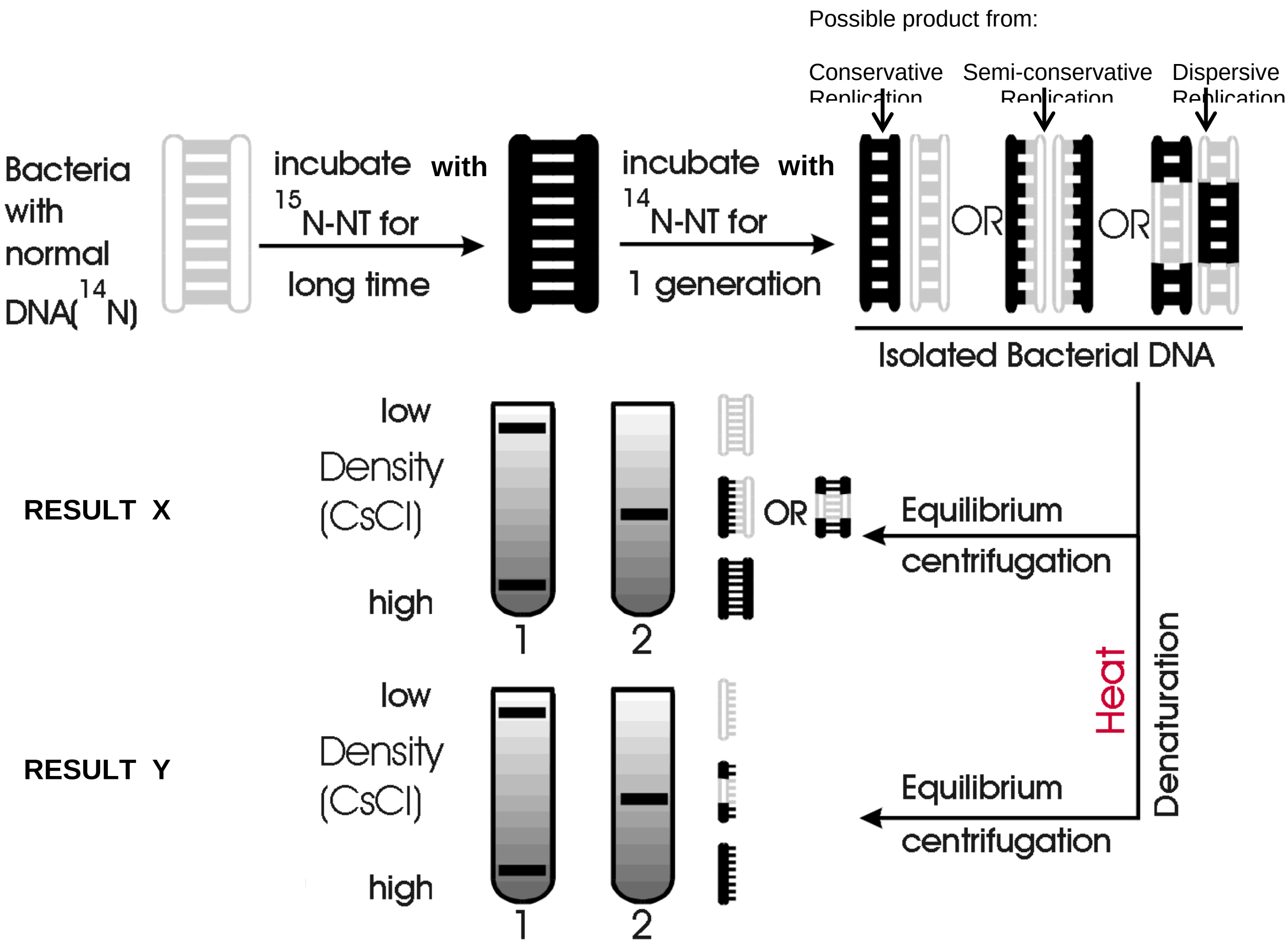


Diagram from: http://www.columbia.edu/cu/biology/courses/c2005 06/lectures/lec11_06.html

Which result (tube 1 or 2) should be obtained, at each centrifugation X and Y, given that replication is semi-conservative? Which type of replication does the result obtained from centrifugation X and Y effectively eliminate?

	Centrifugation X		Centrifugation Y	
	Correct result	**Eliminates**	**Correct result**	**Eliminates**
A	1	Dispersive	1	Conservative
B	2	Conservative	2	Dispersive
C	1	Dispersive	2	Conservative
D	2	Conservative	1	Dispersive

17 *trp*1 to 5 (shaded in diagram) are five genes coding for five different enzymes involved in tryptophan synthesis in yeast cells.

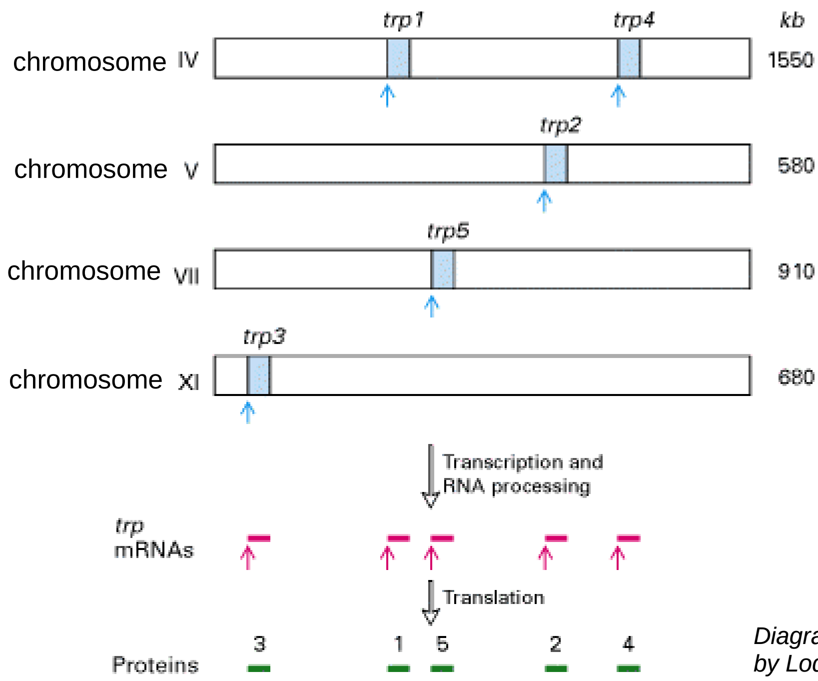


Diagram from "Molecular Biology of the Cell" by Lodish et.al.

What difference could we expect in the organization of the *trp* genes in a bacterial cell?

A Each *trp* gene would be on a different circular chromosome rather than a linear chromosome.

B Each *trp* gene would have their own promoter and terminator.

C *trp* genes would produce polycistronic mRNA.

D *trp* genes would be give rise to more types of mRNA.

18 A New Junior College student just learnt about the fact that mature mRNA have poly-A-tails about 100 nucleotides long.

He reasoned that he should be able to find a sequence of 100 nucleotides with adenine at the end of each eukaryotic gene sequence.

However he was unable to find such sequences of nucleotides when he observed the eukaryotic DNA. Why was this so?

A He should have been looking for multiple nucleotides with thymine instead.

B He was looking at the wrong complementary strand of DNA.

C The DNA had been degraded and so lacked the poly-A-tails.

D Poly-A-tails only exist in mature mRNA but not in DNA.

19 A gene that was 5055 base pairs resulted in the expression of functional proteins that were 350, 450 and 1500 amino acids long. What could be most likely reason for this?

A mRNA degradation

B Gene amplification

C Alternative splicing

D Mutations resulting in early termination signals

20 Repetitive sequences do not have the function of

A coding for functional proteins

B being transcribed into RNA

C binding to specific transcription factors

D being used for DNA fingerprinting

21 Which of the following statements about cancer cells is true?

A They are originally derived from somatic cells

B They are not able to differentiate into other cell types

C Benign tumours have the ability to metastasize

D Increased risk of developing cancer may be due to uv radiation, age and genetic factors

22 Which of the following statements about bacteria is true?

A Bacteria possess 80s ribosomes , introns and RNA processing

B Bacteria possess 70s ribosomes , introns and no RNA processing

C Bacteria possess 70s ribosomes , no introns and no RNA processing

D Bacteria possess 80s ribosomes , RNA processing and no introns

23 In which of the following changes is most energy released during the metabolism of carbohydrates?

A Maltose to glucose

B Glucose to phosphoglyceraldehyde

C Phosphoglyceraldehyde to pyruvic acid

D Pyruvic acid to carbon dioxide and water

24 Which one of the following is a correct outline of the main events in photosynthesis ?

A Light joins carbon dioxide to an acceptor compound which is then reduced by hydrogen obtained from water.

B Light splits water and the resulting hydroxyl group combines with a compound which has incorporated carbon dioxide

C Light splits carbon dioxide and the resulting carbon then combines with oxygen and hydrogen obtained from water.

D Carbon dioxide combines with an acceptor compound and this is reduced by hydrogen split from water by light

25 Fig. 2A and Fig. 2B show the results of two separate experiments on carbon dioxide fixation in photosynthesis using a unicellular green alga and the radioactive isotope ^{14}C to label carbon dioxide. The algae were actively photosynthesizing before the start of both experiments.

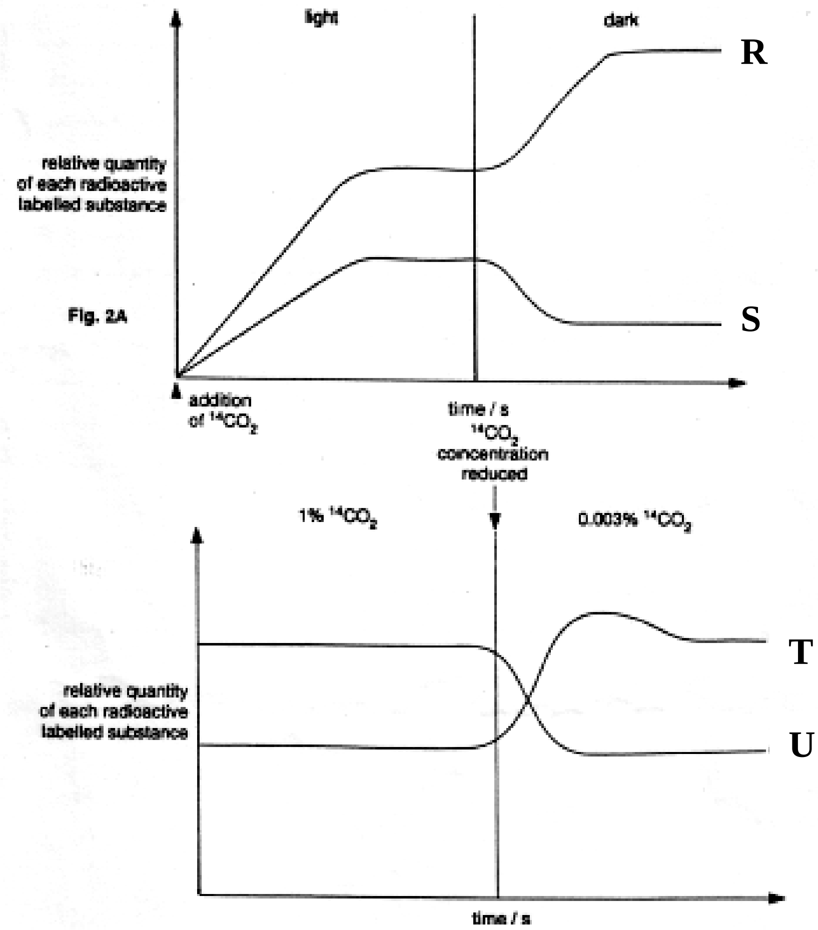


Fig. 2B

With reference to figures 2A and 2B.identify the graphs **R, S, T,** and **U.**

	R	S	T	U
A	RuBP	GP/PGA	GP/PGA	RuBP
B	GP/PGA	RuBP	GP/PGA	RuBP
C	GP/PGA	RuBP	RuBP	GP/PGA
D	RuBP	GP/PGA	RuBP	GP/PGA

26 What is not necessary for PCR to occur?

A dATP

B Primers

C DNA fragments

D Ribonucleotides

27 Anti-thrombin (a protein involved in blood clotting) and human growth hormone are examples of human proteins that can be produced via the genetic engineering technique.

How would the process which produces anti-thrombin differ from the one that produces a bacterial antibiotic resistance protein if both were to be carried out in bacteria?

A Antibiotic resistance would not be used as a selection marker for successfully transformed bacteria for the process that produces the antibiotic resistance protein.

B The Anti-thrombin gene insert would require an additional eukaryotic promoter.

C Anti-thrombin would require additional post-translational modification.

D Different plasmids would have to be used.

28 In parallel with the Human Genome Project, the DNA of a set of model organisms (such as bacteria, roundworms, fruitflies, mice and chimpanzees) are being studied.

Which technique might be most useful in using the information from model organisms to provide critical clues about the structure and function of human genes?

A Sequencing

B DNA fingerprinting

C Comparing molecular homology

D Finding the regions of repetitive sequences

29 The genetically engineered supersalmon was created from atlantic salmon stocks and are capable of growing to a large size in 14 months. Which of the following is not a benefit intended from the crop?

A Higher yield for farmers

B Minimising pollution

C Decreasing the food consumption of the crops in their lifetime

D Increase in supply to meet world's demand

30

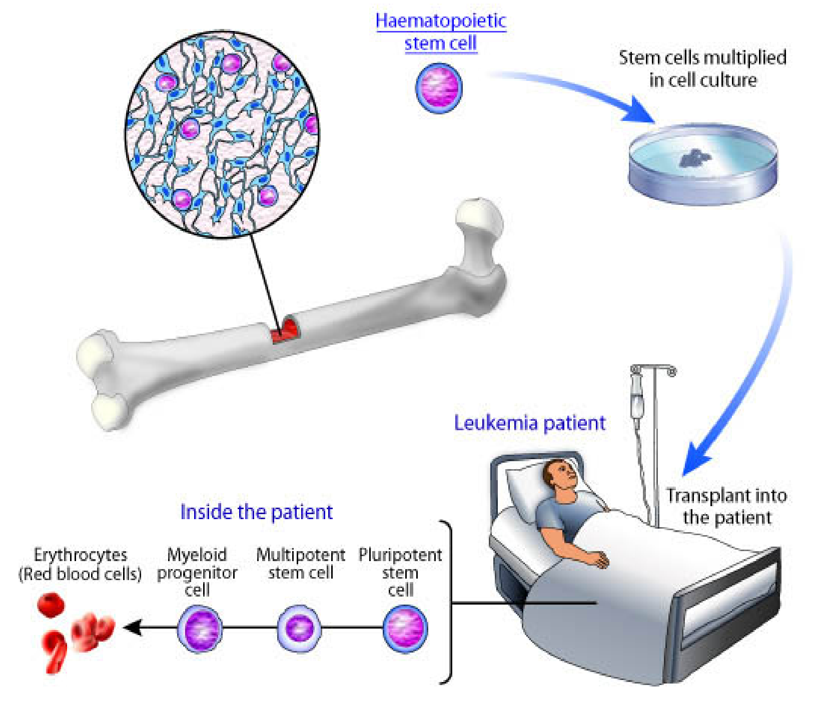


Illustration by Cell Imaging Core of the Center for Reproductive Sciences.

Of the following, which of the following options is not a characteristic of the form of treatment shown above?

i. Easy to extract in laboratory
ii. Readily available
iii. Difficult to control
iv. Limited longevity

A i only **B** i and ii **C** i and iii **D** i, ii, and iii

--- End of Paper ---